Industrial Internship Report on

"Crop and weed detection"

Prepared by

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Executive Summary
ThisAgriculture plays an important role in food production, and weed management is one of the major challenges faced by farmers. Weeds compete with crops for essential resources such as water, nutrients, sunlight, and space, which can reduce crop productivity. Traditional weed-control methods often involve spraying pesticides or herbicides uniformly across agricultural fields, which can result in unnecessary chemical usage and may affect crops and the environment. The Crop and Weed Detection project aims to develop an intelligent object detection system capable of identifying and distinguishing crops and weeds in agricultural images. The project uses the YOLOv8 deep-learning object detection model to detect crops and weeds by drawing bounding boxes around them. The dataset consists of approximately 1,300 images with corresponding YOLO-format annotation files. The dataset contains two classes: **crop** and **weed**. The data was prepared and divided into training, validation, and testing sets before being used for model training. The trained model can be used as the foundation for a precision agriculture system in which pesticides or herbicides are applied specifically to detected weeds rather than being sprayed uniformly across the entire field.

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1 Preface

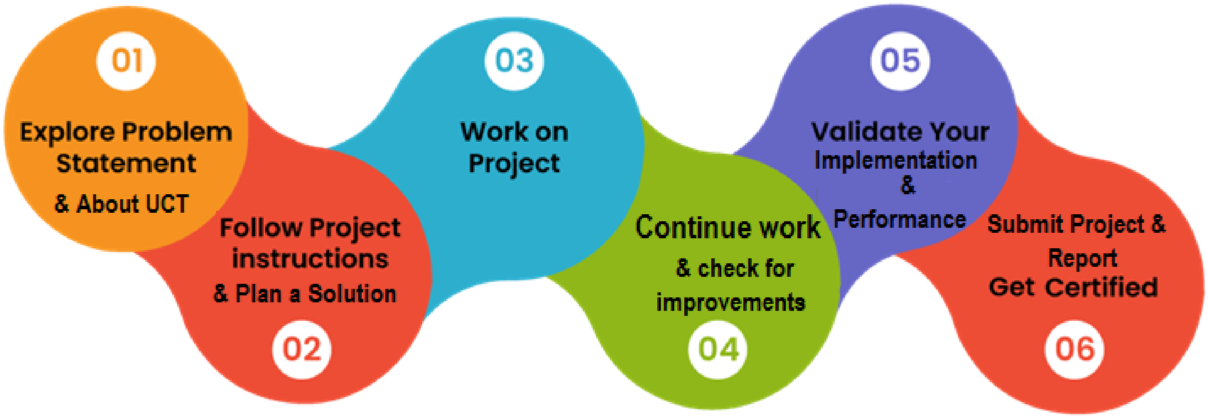
During this internship in **Data Science and Machine Learning** at **upSkill Campus**, I worked on a real-world project titled **Crop and Weed Detection Using YOLOv8**. Throughout the internship, I learned the complete machine learning workflow, including understanding the problem statement, preparing datasets, data preprocessing, model training, evaluation, deployment, and documentation. I also gained practical experience in computer vision using the YOLOv8 object detection model and developed a Flask-based web application for crop and weed detection.

Internships play an important role in bridging the gap between academic learning and industry requirements. This internship provided me with practical exposure to machine learning, deep learning, computer vision, and web application development. It helped me understand real-world project development, improve my technical skills, and gain confidence in applying theoretical knowledge to solve practical problems.

My project, **Crop and Weed Detection Using YOLOv8**, focuses on identifying crops and weeds in agricultural images using deep learning. Weeds compete with crops for nutrients, water, sunlight, and land, which negatively affects crop production. The objective of this project is to accurately detect crops and weeds so that pesticides can be applied only where weeds are present. This supports precision agriculture by reducing unnecessary pesticide usage and improving farming efficiency.

I sincerely thank **upSkill Campus** and **Uniconverge Technologies (UCT)** for providing me with this valuable internship opportunity. The internship allowed me to gain practical knowledge in Data Science and Machine Learning while working on an industry-oriented project. It also enhanced my understanding of project development, problem-solving, teamwork, documentation, and GitHub project management.

The internship was conducted in a structured manner over six weeks. Initially, the project problem statement and dataset were introduced. The following weeks involved dataset preparation, preprocessing, YOLOv8 model training, model evaluation, Flask web application development, testing, report preparation, and GitHub repository creation. Weekly progress reports were submitted to monitor project development and learning throughout the internship.



This internship greatly improved my understanding of machine learning and computer vision concepts. I learned how to prepare datasets, train and evaluate YOLOv8 models, deploy a Flask application, use Git and GitHub for version control, and document a complete project professionally. Overall, this internship strengthened my technical skills, enhanced my confidence, and provided valuable hands-on industry experience.

I would like to express my sincere gratitude to the **upSkill Campus Team**, **Uniconverge Technologies (UCT)**, the internship mentors, and all the instructors for their valuable guidance and support throughout the internship.

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2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

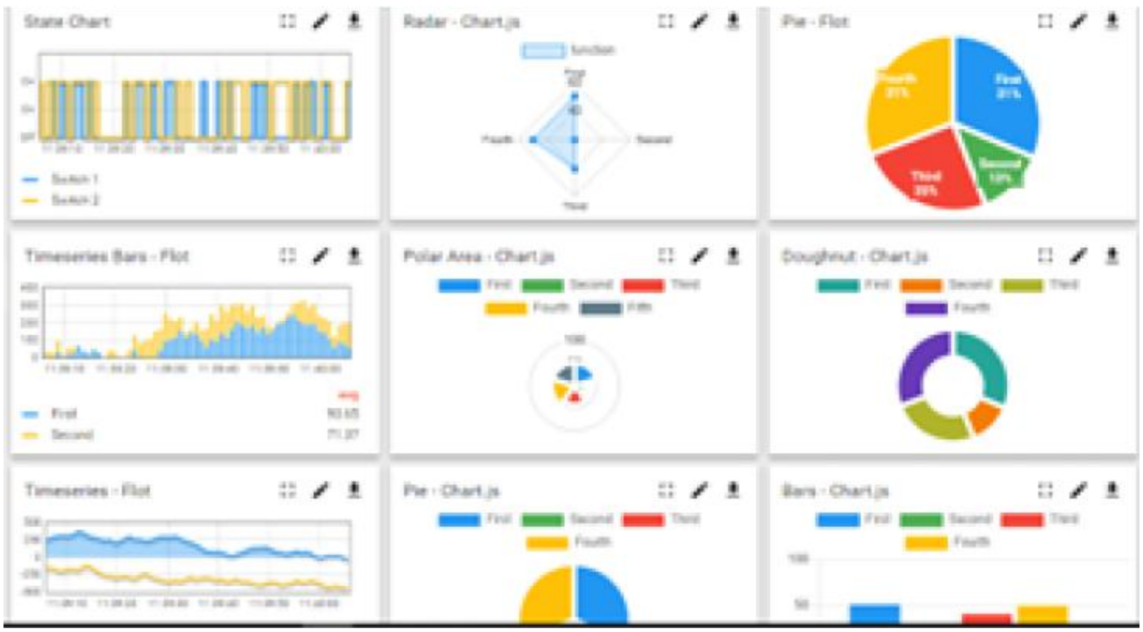
i. UCT IoT Platform ( )

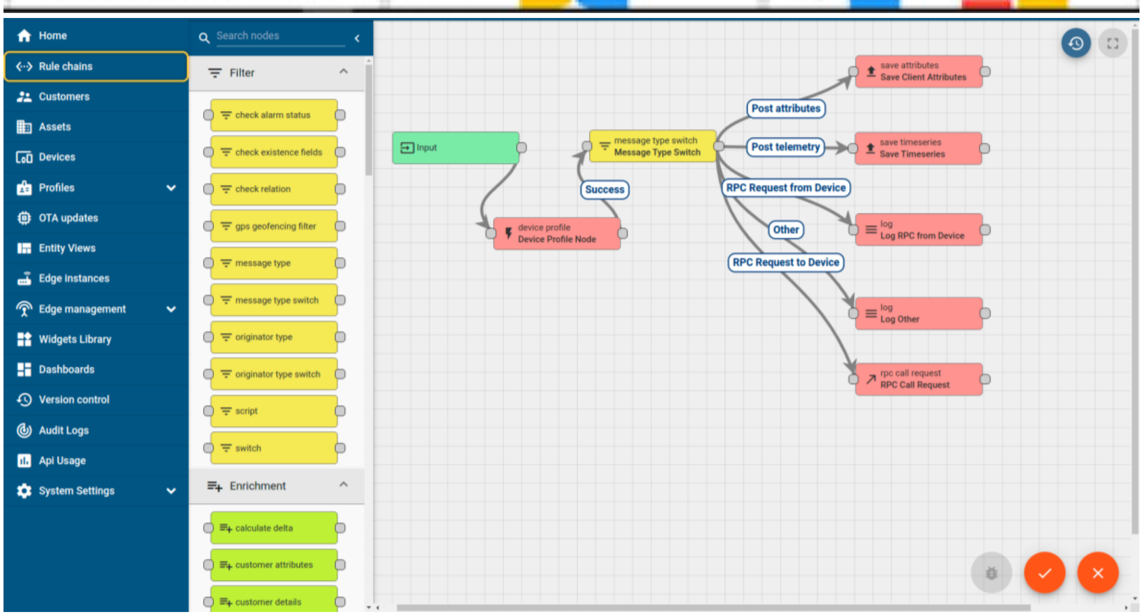
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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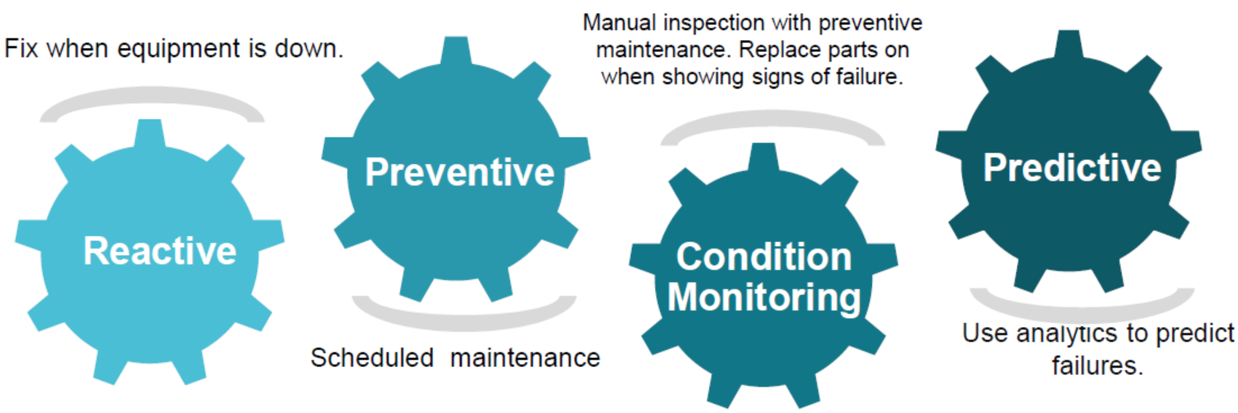
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

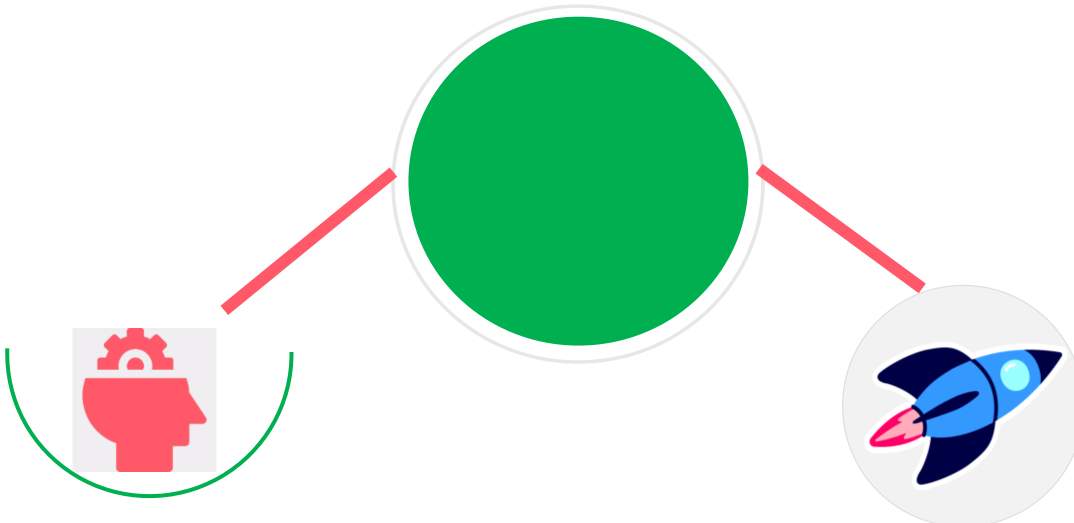
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

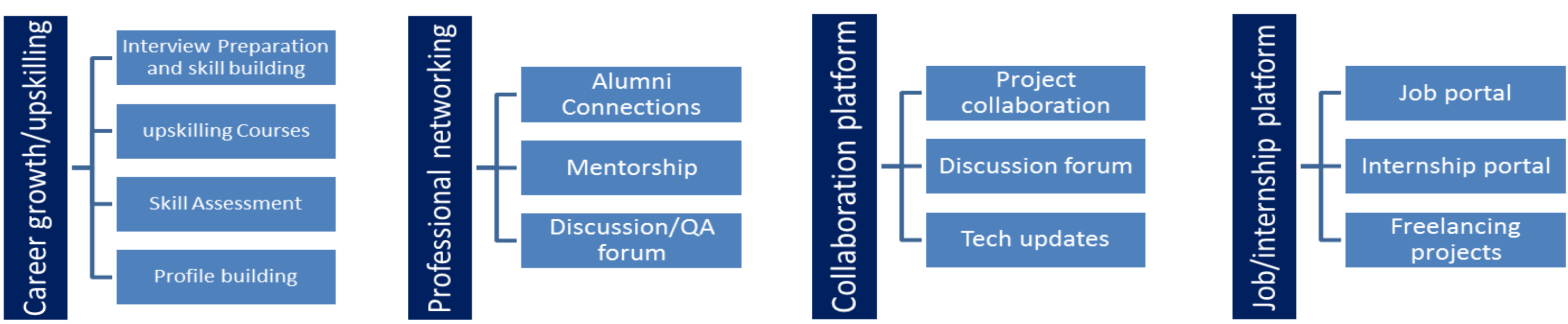
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ to gain practical experience in Data Science and Machine Learning

☛ to understand the complete machine learning project lifecycle.

☛ to develop problem-solving and analytical skills through a real-world project

☛ to improve technical knowledge in Python, Computer Vision, and Deep Learning.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] https://www.ultralytics.com/

[2] https://docs.ultralytics.com/

[3] https://github.com/ultralytics/ultralytics

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Machine Learning	ML
Computer Vision	CV
You Only Look Once	YOLO
Internet of Things	IoT

3 Problem Statement

In the assigned problem statement

Weeds are unwanted plants that grow alongside agricultural crops and compete with them for nutrients, water, sunlight, and land. Their presence can negatively affect crop growth and reduce agricultural productivity.

Traditional weed management methods may involve manual removal or widespread application of pesticides and herbicides. Uniform spraying can lead to excessive chemical usage, increased costs, environmental concerns, and unnecessary exposure of crops to chemicals.

Therefore, there is a need for an automated system capable of distinguishing between crops and weeds. The objective of this project is to use computer vision and deep learning to automatically detect crops and weeds in images. The detected weed locations could eventually be used by smart agricultural equipment to selectively spray chemicals only where weeds are present.

4 Existing and Proposed solution

Traditional approaches to weed management include:

- Manual identification and removal of weeds.
- Mechanical weed-removal techniques.
- Uniform spraying of pesticides or herbicides across agricultural fields.
- Manual inspection of fields by farmers.

These methods can be time-consuming and labor-intensive. Uniform chemical spraying may also result in unnecessary pesticide usage because chemicals are applied even in areas where weeds are not present.

Some modern agricultural systems use image processing and computer vision techniques. However, conventional image-processing methods may have difficulty handling variations in lighting, plant appearance, background conditions, and overlapping vegetation.

The proposed solution is an AI-based Crop and Weed Detection System using **YOLOv8**.

The system processes agricultural images and identifies two object classes:

- **Crop**
- **Weed**

The workflow consists of:

Input Image → Data Preprocessing → YOLOv8 Model → Object Detection → Crop/Weed Classification → Bounding Box Visualization → Detection Result

The dataset is first validated and organized into training, validation, and testing sets. A pretrained YOLOv8 model is then fine-tuned using the custom crop-and-weed dataset.

After training, the model can process new agricultural images and predict the locations of crops and weeds using bounding boxes and class labels.

A Flask-based web application is also used to provide a simple interface where users can upload an image and view the detection results.

The proposed system provides several potential benefits:

- Automated identification of crops and weeds.
- Reduced dependence on manual field inspection.
- Support for precision agriculture.
- Potential reduction in unnecessary pesticide or herbicide usage.
- Faster processing of agricultural images.
- Ability to identify multiple objects in a single image.
- Potential integration with smart spraying systems, agricultural robots, and drones.

4.1 Code submission (Github link)

https://github.com/Rahitya28/upskillcampus

4.2 Report submission (Github link):

https://github.com/Rahitya28/upskillcampus/blob/main/CropAndWeedDetection_Rahitya_USC_UCT.pdf

5 Proposed Design/ Model

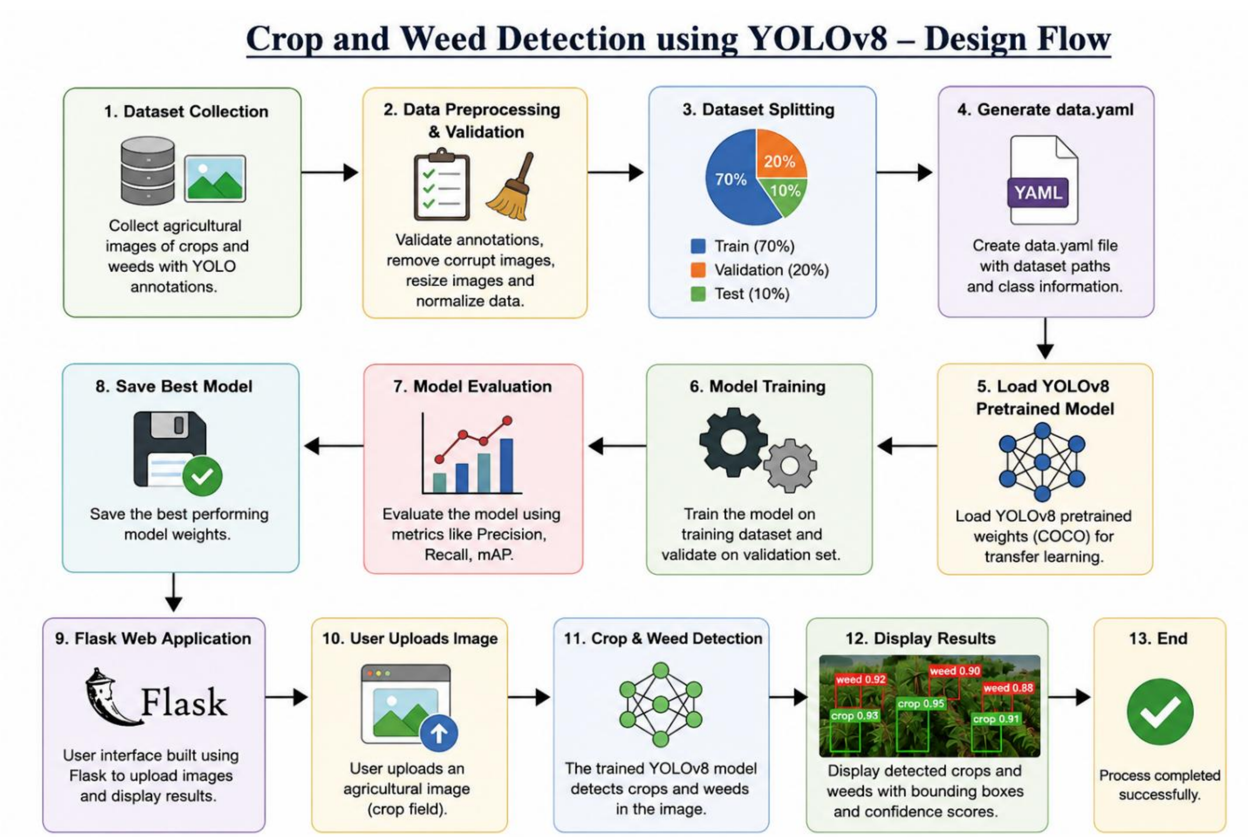
The proposed system is designed to automatically detect and classify crops and weeds in agricultural images using the YOLOv8 object detection model. The overall workflow consists of data collection, preprocessing, model training, prediction, and result visualization.

The design begins with collecting the agricultural image dataset containing crop and weed images with YOLO-format annotations. The dataset is validated, cleaned, and organized into training, validation, and testing sets. A data.yaml configuration file is then generated for YOLOv8 training.

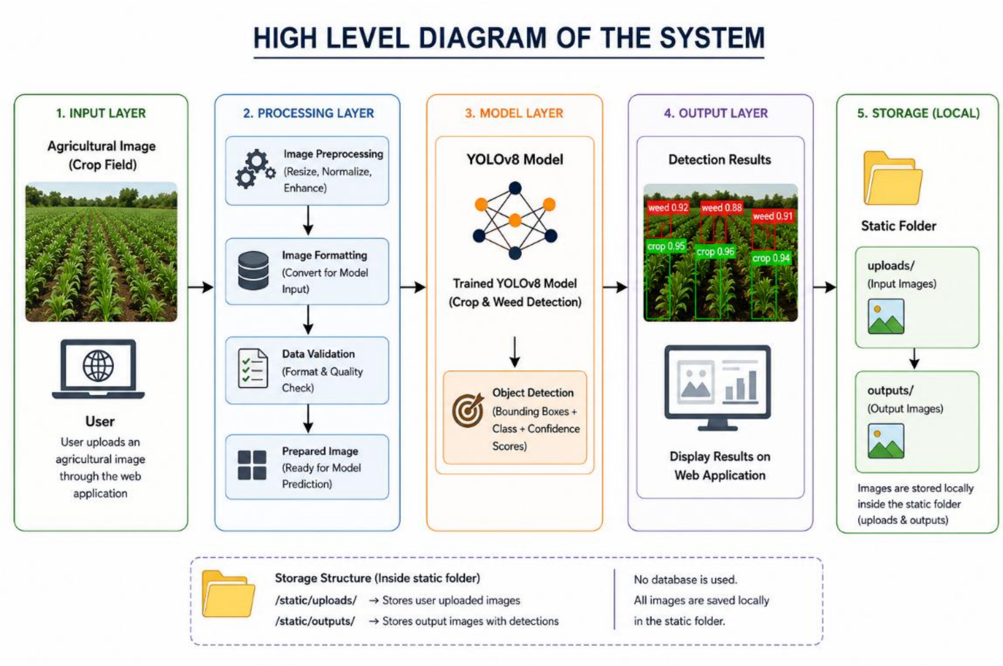
A pretrained YOLOv8 model is fine-tuned using the prepared dataset. During training, the model learns to identify the location and class of each object by predicting bounding boxes and confidence scores.

After training, the best-performing model is used for inference. Users can upload an agricultural image through the Flask web application, where the trained model processes the image and detects crops and weeds. The final output is displayed with labeled bounding boxes and confidence scores.

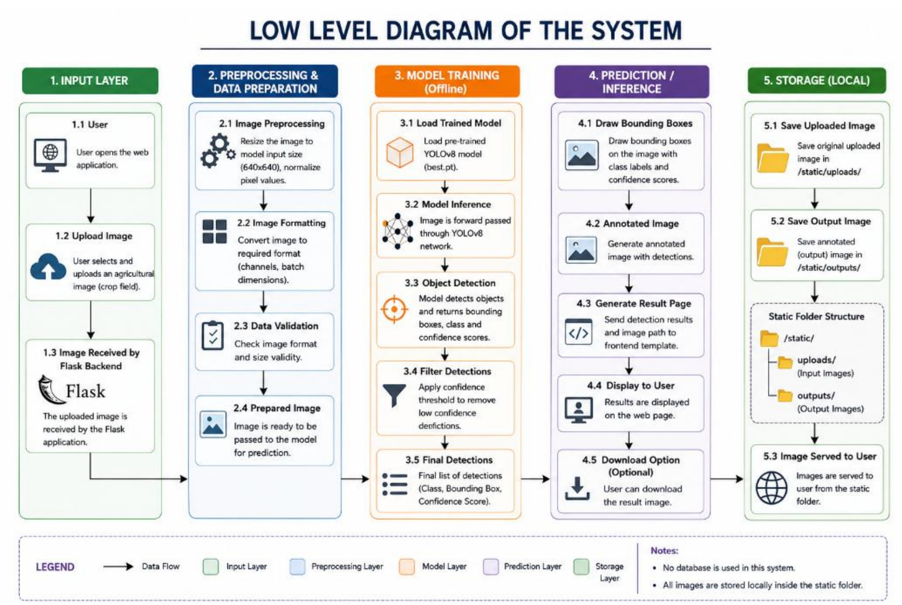
This design provides an efficient and scalable solution for precision agriculture by enabling selective weed detection, which can help reduce unnecessary pesticide usage.



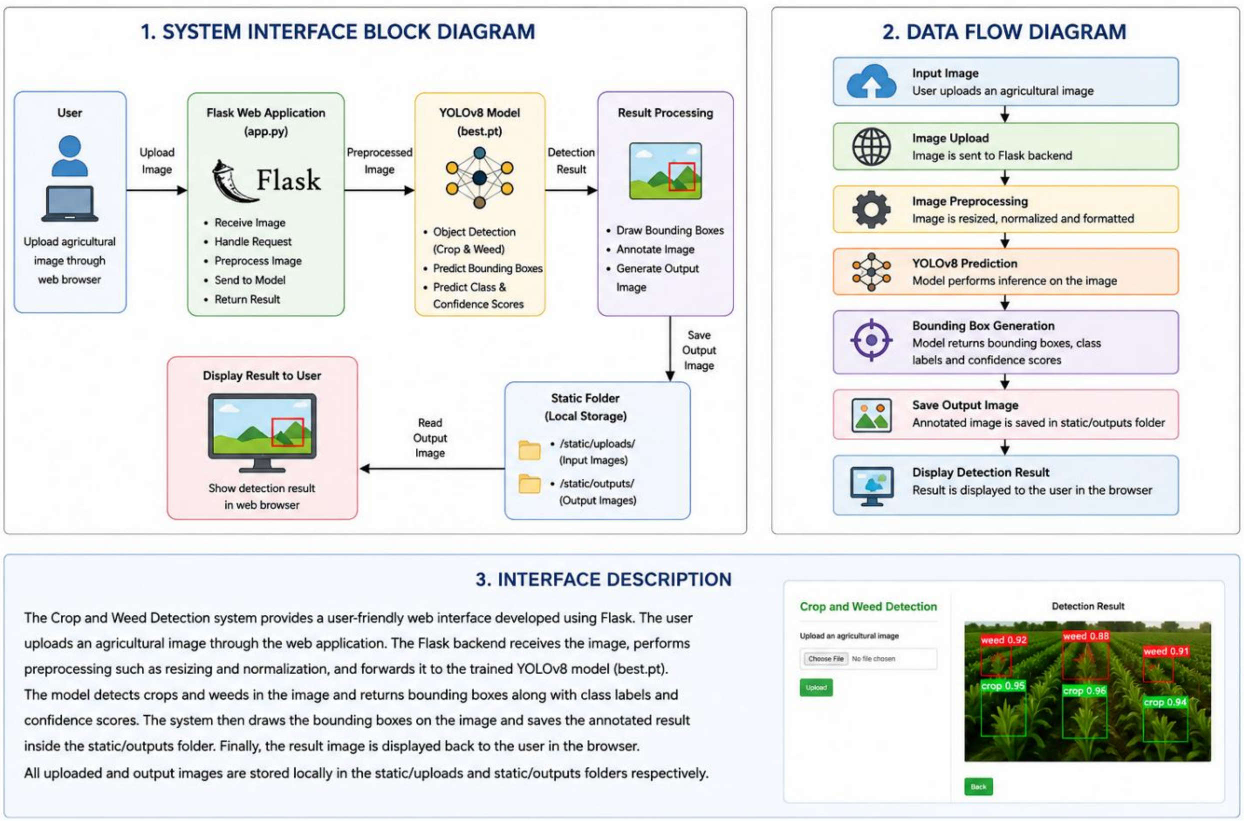
5.1 High Level Diagram (if applicable)



5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)



6 Performance Test

The performance of the proposed Crop and Weed Detection system was evaluated based on detection accuracy, inference speed, memory utilization, and usability. The system was tested using unseen agricultural images containing both crops and weeds. The trained YOLOv8 model successfully detected and classified the majority of objects with high confidence while maintaining fast prediction time.

The major constraints considered during development were model accuracy, inference speed, image quality, and hardware limitations. Image preprocessing and the use of the lightweight YOLOv8 model helped improve performance while reducing computational requirements.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Output	Result
TC-01	Upload valid crop image	Image uploaded successfully	Pass
TC-02	Upload image containing weeds	Weed detected with bounding box	Pass
TC-03	Upload image containing crops	Crop detected correctly	Pass
TC-04	Upload image containing both crop and weed	Both objects detected separately	Pass
TC-05	Upload unsupported file	Error message displayed	Pass
TC-06	Display prediction result	Annotated image shown in browser	Pass

6.2 Test Procedure

- Launch the Flask web application.
- Upload an agricultural image through the browser.
- The application stores the uploaded image in the static/uploads folder.
- The image is preprocessed and passed to the trained YOLOv8 model.
- The model predicts crops and weeds with bounding boxes and confidence scores.
- The annotated image is saved in the static/outputs folder.
- The output image is displayed on the web page.
- Repeat the process using different test images to verify consistent performance.

6.3 Performance Outcome

Performance Metric	**Result**
Model Used	YOLOv8n
Classes	Crop, Weed
Image Size	640 × 640
Detection Accuracy	High for most test images
Inference Time	Approximately 0.05–0.15 seconds per image (depending on hardware)
Output	Bounding boxes with class labels and confidence scores
Storage	Images stored locally in static/uploads and static/outputs
User Interface	Flask Web Application

7 My learnings

During this internship, I gained valuable hands-on experience in Data Science, Machine Learning, and Computer Vision by developing a Crop and Weed Detection system using the YOLOv8 object detection model.

I learned the complete machine learning project lifecycle, including dataset preparation, image preprocessing, data splitting, model training, evaluation, and deployment. I also understood the importance of data quality and annotation in improving model performance.

Through this project, I enhanced my Python programming skills and gained practical experience with Flask for developing a web application. I also learned how to integrate a trained deep learning model into a user-friendly interface for real-time prediction.

Additionally, I improved my knowledge of Git and GitHub by managing project files, creating repositories, and documenting my work professionally. Preparing reports, testing the application, and troubleshooting errors helped strengthen my problem-solving, analytical thinking, and project management skills.

Overall, this internship provided practical exposure to real-world AI applications in agriculture and significantly improved my confidence in developing and deploying machine learning solutions. The knowledge and experience gained will be highly beneficial for my future career in Data Science, Machine Learning, and Artificial Intelligence.

8 Future work scope

Although the developed system successfully detects crops and weeds using the YOLOv8 model, there are several opportunities for future enhancements.

- Increase the dataset size by collecting more agricultural images from different environments and lighting conditions.
- Extend the model to detect multiple crop and weed species instead of only two classes.
- Improve model accuracy by experimenting with larger YOLO models and advanced data augmentation techniques.
- Develop a real-time detection system using live camera or drone footage for smart farming applications.
- Integrate GPS and IoT sensors to provide location-based weed detection and precision agriculture support.
- Deploy the application on cloud platforms to enable remote access and large-scale usage.
- Develop a mobile application that allows farmers to capture images directly from smartphones and receive instant detection results.
- Integrate an automated pesticide spraying mechanism that selectively sprays weeds based on the detection results, reducing pesticide usage and improving crop yield.